

Employment

Graduate Research Assistant	NC State University	Apr 2025 – Present
- Collaborated with plant biologists and geneticists to deploy cloud-native genomic data platform on GCP (Cloud Run) with JBrowse 2 genome browser, enabling 100+ researchers to visualize 500GB+ genome datasets. - Improved ML model accuracy by 12% via hyperparameter tuning and time-series analysis on biological sensor data.		
Product Engineer Intern	Isomer AI	May 2025 – May 2026
- Collaborated with product and design teams to ship 10+ production AI features: multi-model LLM pipelines (Claude, GPT-4o, Gemini), Temporal.io agentic workflows; 923 commits, 7 production deployments. - Owned full-stack React/Node.js + TypeScript codebase: PostgreSQL, Redis, Docker, Terraform, Auth0 RBAC, and Playwright test coverage.		
Software Developer Intern	Mphasis (HP Company)	Jun 2024 – Aug 2024
- Partnered with healthcare data teams to build 5+ AI workflows including health claim automation using OCR, NLP, and ML pipelines; improved document processing accuracy by 25% across 10K+ daily medical claims. - Automated AWS and GCP CI/CD pipelines; integrated Grafana monitoring and led model validation and red-teaming sessions.		
AI/ML Intern	Capgemini	Jun 2023 – Aug 2023
- Led cross-functional SDLC for data-driven enterprise ticket analysis using NLP classification models; worked with support teams to reduce resolution time by 15% and eliminate 2,000+ monthly manual triages. - Built Kafka + Spark + Hadoop pipelines with data engineers; cut ticket-processing latency by 20% at peak load.		

Education

Raleigh, NC	North Carolina State University	Aug 2024 – May 2026
- Master of Science in Computer Science - Courses: Artificial Intelligence, Operating Systems, Algorithms, Data Science, Object Oriented Development, Software Engineering, Programming Languages, Computer Networks, Distributed Systems.		
Mumbai, India	Mumbai University	Aug 2020 – May 2024
Bachelor of Technology in Computer Science; CGPA: 9.32/10. Courses: Data Warehousing, Big Data Infrastructure, DS & Algorithms, Machine Learning, Blockchain, Computer Networks, Database Systems.		

Projects

AlgoPulse	Full-Stack Agentic AI Platform	2025
- Built agentic AI orchestration platform on Temporal.io and Kafka supporting multi-step reasoning pipelines; sustained 2,000 async inference requests per second under sustained production load. - Replaced polling with event-driven Supabase Edge Functions; sustained sub-200ms SSE delivery at 2,000 req/sec.		
CortexQ	Kubernetes LLM Inference Platform	2026
- Built Kubernetes-native LLM inference platform: FastAPI router, KEDA autoscaler (scale-to-zero to N replicas), Kopf CRD operator for multi-provider LLM serving; circuit breaker + EWMA load balancing. - Deployed via Helm + ArgoCD GitOps on Kubernetes; OTel + Prometheus + Grafana observability; 88 pytest tests.		
PulseRank	Real-Time Leaderboard & Activity Feed	2026
- Built event-driven leaderboard and activity feed in TypeScript with NestJS and Next.js; MySQL transactional writes, MongoDB feed projections with TTL, Redis sorted sets for sub-millisecond reads; live SSE. - Validated multi-DB consistency with Testcontainers; benchmarked hot paths with k6 at 10K concurrent users.		

Languages and Technologies

- Languages:** TypeScript, JavaScript, Python, Java, SQL, Bash
- Full Stack:** React, Next.js, NestJS, Node.js, PostgreSQL, Redis, MongoDB, MySQL, REST APIs
- Infra & Cloud:** Kubernetes, Docker, Terraform, GCP, AWS, GitHub Actions, Helm, ArgoCD, Kafka
- AI & Observability:** LangChain, Temporal.io, Claude API, GPT-4o, OpenTelemetry, Prometheus, Grafana, Auth0

Additional Experience and Awards

- IEEE Published:** Co-authored and presented research on Computer Vision-Based Cybersecurity Threat Detection at the Springer icSoftComp 2023 international conference, reaching an audience of 500+ global attendees.
- Hackathon Winner:** Won or ranked Top 10 in 6+ national hackathons; built 3 full-stack MVPs end-to-end in 48-hour competitive sprints.
- GDSC Lead:** Led 4+ tech events as Google Developer Student Club Lead, reaching 500+ students across hackathons and conferences.
- Technical Writing:** Published articles on LLM model collapse, AI agent security, and audio AI at kulkarniaditya.com/blog.